

MTH-113-2.00 - Introduction to Quantitative Thinking

Course Name	MTH-113-2.00	Course Title	Introduction to Quantitative Thinking
PS Course ID	669447	Base Program	FXCRD-1.00
Course Version	2.00	Credits	5
Academic Year Version First Offered	2025	School	School of Arts and Sciences
Academic Year Version Last Offered		Primary Course Author	Achyut Kumar Banerjee
Course Status	Active	Course Review Status	Approved
Availability Outside Base Program	Yes	Base Program for Requested Course	Flexible credit
Requested Course Name	Quantitative Reasoning: Introduction	Total Assessment Percentage	100
Curricular Group Information			
Curricular Group 1	Flexible Credits	Curricular Group 2	
Curricular Group 3		Curricular Group 4	
Curricular Group 5			
Course Type Information			
Course Type 1	In person on campus	Course Type 2	
Course Type 3		Course Type 4	Classroom instruction
Course Type 5			
Pre-requisite and Eligibility			
Prerequisites Course 1		Prerequisites Course 2	
Prerequisites Course 3		Prerequisites Course 4	
Prerequisites Course 5		Eligibility Criteria	
Any Other Exclusion			
Other Details			
Course delivery mode	Offline	Level	1
Number Of Teaching Hours	60		
UDL Details			
Alignment to UDL	No		
Description of UDL Alignment			
UDL Alignment	Click here to view the uploaded document !!		
Development Team			
Author			
Author's Name		Role	
Achyut Kumar Banerjee		Author	
Co-Author			
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Abstract Description

This foundational course aims to introduce students to the core components of quantitative reasoning and mathematical methods, equip them with practical data skills, including the ability to interpret graphs, calculate statistical measures, and use spreadsheet software effectively to solve problems. The course does not have any prerequisites; rather, it intends to impart foundational knowledge of mathematical thinking that is expected to help students in other courses across disciplines.

Description

Introduction to Quantitative Thinking is a six-week summer bridge course designed exclusively for first-year undergraduate students. This support course meets the critical need for mathematical preparation, enabling all students, regardless of their prior mathematical background, to engage successfully with the core and the common curriculum courses and choose any academic pathway they wish to pursue.

The course will be delivered over six weeks during the summer program, immediately before the first semester begins. Total instruction consists of 60 hours of intensive classroom teaching conducted over four days each week, allowing for personalized support and peer collaboration. This concentrated, immersive format ensures that students build competency rapidly while remaining engaged and supported throughout their learning journey.

Rationale and Introduction

Diversity and inclusion are central to our university. For many of our students, a lack of preparation (especially in Mathematics and English) has been an obstacle to success. This course, along with the language course, is a formalization of the support system aimed at equipping our students with the necessary quantitative skills to navigate the core and common curriculum courses.

As part of the common curriculum in the undergraduate program, every student has to take the courses Public Reasoning (PRE) and The Measures of India (Understanding India-III). The former course requires some experience in logical reasoning, and the latter requires some statistical skills. In addition, it would be ideal to empower our students with the capability to choose any occupational track, including ‘Data, Development, and Democracy’, which requires solid quantitative understanding.

This is, therefore, a support course designed for students who may benefit from additional assistance in developing their foundational knowledge in mathematics and statistics, particularly those who may not have yet fully demonstrated proficiency during the admission process, to successfully navigate the core and common curriculum courses with mathematics and statistics content.

Intended Learning Outcomes	
Intended Learning Outcome	Description
Demonstrate	Demonstrate foundational knowledge of arithmetic operations, exponents, algebraic expressions, and coordinate geometry
Engage	Engage in mathematical thinking and collaborative problem-solving to build confidence and quantitative literacy.
Execute	Execute statistical and probabilistic reasoning, estimation methods, and data visualization to solve real world problems.
Perform	Carry out mathematical operations and problem-solving techniques in numerical, spatial, and data analysis contexts.

Pedagogy	
Pedagogical Practice	Pedagogy Description
Lecture	Concise presentations followed by guided practice and problem-solving to reinforce learning.
Problem based learning	Students solve carefully designed problems individually and in groups with instructor support.
Project-based learning	Students apply multi-unit concepts through real-world collaborative projects involving data management, visualization, and interpretation.

Assessment			
Total Assessment Percentage			100
Assessment Type	Number of ILO Linkage	Assessment Percentage	Description
In-class exam	2	15	Exam 1 (60 minutes duration, to be submitted individually), to be conducted in 2nd week, will be a pen and paper-based assessment on the syllabus of unit 1. The assessment will include a mix of question types (multiple-choice and short problem-solving questions) and will evaluate students' foundational understanding of the quantitative methods.
In-class exam	2	15	Exam 2 (60 minutes duration, to be submitted individually), to be conducted in 3rd week, will be a pen and paper-based assessment on the syllabus of unit 2. The assessment will include a mix of question types (multiple-choice and short problem-solving questions) and will evaluate students' ability to understand algebra and coordinate geometry.
In-class exam	1	15	Exam 3 (60 minutes duration, to be submitted individually), to be conducted in 4th week, will be a computer-based assessment on the syllabus of Unit 3. The assessment will evaluate students' ability to estimate statistical measures in a spreadsheet program.
In-class exam	2	15	Exam 4 (60 minutes duration, to be submitted individually), to be conducted in 6th week, will be a pen and paper-based assessment on the syllabus of unit 4. The assessment will include a mix of question types (multiple-choice and short problem-solving questions) and will evaluate students' ability to apply quantitative knowledge in everyday life.
			The students will be divided into groups. Each group will work on a project related to an everyday life issue (for example, breakfast habit, reading time). This will involve data acquisition (from small surveys), processing, and visualization. The project will start in the 3rd week of the course. Each group will present their project findings as a digital/poster presentation. The presentation will carry 10% weightage. It will be evaluated based

Project	2	30	on topic selection and presentation skill (including visual appeal and interaction). Individual members must indicate their roles in the project. Individual student will submit a brief report, which will contribute 10% to the overall grade. The report will be evaluated based on its organization and information of the workflow (data collection, analysis, and interpretation of the findings). The viva-voce will evaluate contribution and depth of knowledge of individual group members and will contribute 10% to the total grade. In the viva-voce, the instructor may ask the students to demonstrate use of a specific tool or function of the software related to their projects. The instructor should ensure that the groups are heterogeneous with students across all disciplines. The teams should not be larger than 5 and not smaller than 2. All projects will be discussed a priori. The students should start working on the projects from week 3. The presentation, report submission, and viva-voce will be conducted in week 5.
Class participation	4	10	This will be based on the students' interactions with peers and their contribution to discussions in the class. The students will be divided into groups and engaged in different types of in-class activities. The instructor may use this assessment to gather continuous feedback from the students, including their challenges in navigating the course, and take the necessary actions.

Syllabus Summary

The course is structured into four progressive units that build quantitative reasoning from fundamentals to application.

- Unit 1 establishes essential building blocks through arithmetic operations, factorization, exponents and powers, and the unitary method for measurement. (Week 1)
- Unit 2 develops algebraic thinking and spatial representation through linear, polynomial, and quadratic relationships, and their graphical representations using coordinate geometry. (Weeks 2-3)
- Unit 3 builds data competency through an introduction to statistics, data visualization, interpretation, and understanding data distributions. (Weeks 3-4)
- Unit 4 extends quantitative reasoning to numerical applications, including percentage and compound interest, ratio and proportion, permutations and combinations, sequences and series, and probability. (Weeks 5-6)

The key resources for this course are listed below:

Textbooks:

1. [Textbooks \(VI – XI\) by NCERT](#)
2. NCERT Exemplar Problems

Syllabus

Unit Name

Unit 1: Foundation for Quantitative Methods

Description

Position the unit to build essential building blocks for more advanced topics.

Week 1:

- 1.1 Number System
- 1.2 Arithmetic Operations
- 1.3 Prime Factorization (LCM and HCF)
- 1.4 Exponents and Power
- 1.5 Unitary Method

Readings

Core Readings:

NCERT Class VI, Chapters:

- Playing with Numbers
- Integers
- Fractions
- Decimals

NCERT Class VII, Chapters:

- Integers
- Fractions and Decimals
- Exponents and Powers

NCERT Class VIII, Chapter:

- Exponents and Powers

All core readings for this chapter are uploaded [here](#).

Supplementary Readings:

NCERT Class VII, Chapter: Rational Numbers

Specifics Units of:

[NCERT Exemplar Class VI](#)

[NCERT Exemplar Class VII](#)

[NCERT Exemplar Class VIII](#)

Unit Name

Unit 2: Algebra and Coordinate Geometry

Description

Algebra is a useful tool to convert the problem into solving a system of linear equations. Solutions to linear equations also arise in several real-life applications. Graphs of linear functions will also be discussed.

Week 2:

2.1 Introduction to Algebra

2.2 Algebraic Expression

2.3 Factorization

2.4 Algebraic Identities

2.5 Polynomials

Week 3:

2.6 Linear Equations

2.7 Quadratic Equation

2.8 Coordinate Geometry

2.9 Graphs of Linear Equations

Readings**Core Readings:**

NCERT Class VI, Chapter:

- Algebra

NCERT Class VII, Chapters:

- Algebraic Expression
- Simple Equation

NCERT Class VIII, Chapters:

- Algebraic Expression and Identities
- Factorization
- Linear Equation in One Variable

NCERT Class IX, Chapters:

- Linear Equation in Two Variables
- Polynomials
- Coordinate Geometry

NCERT Class X, Chapters:

- Linear Equation in Two Variables
- Polynomials
- Quadratic Equations

All core readings for this chapter are uploaded [here](#).

Supplementary Readings:

Specific Units of:

[NCERT Exemplar Class VI](#)

[NCERT Exemplar Class VII](#)

[NCERT Exemplar Class VIII](#)

[NCERT Exemplar Class IX](#)[NCERT Exemplar Class X](#)**Unit Name**

Unit 3: Statistics, Data Management and Visualization

Description

Plotting and interpreting graphs is of considerable importance as a life skill. Line graphs, pie and bar charts, organization of quantitative data, measures of central tendency, and measures of variance will be covered in this unit. Students will also be introduced to the use of software (MS Excel or Google Sheets). They will be required to use the software to enter simple formulas and use it to calculate statistical measures.

Week 3:

3.1 Introduction to Statistics

i) Measures of Central Tendency

ii) Measures of Variance

Week 4:

3.2 Normal Distribution

3.3 Data Handling and Plotting Graphs

3.4 Data Visualization and Interpretation using Google Spreadsheet or MS Excel

Readings**Core Readings:**

NCERT Class VII, Chapter:

- Data Handling

NCERT Class VIII, Chapter:

- Introduction to Graphs

NCERT Class IX, Chapter:

- Statistics

NCERT Class X, Chapter:

- Statistics

NCERT Class XI, Chapter:

- Statistics

All core readings for this chapter are uploaded [here](#).**Supplementary Readings:**[NCERT Exemplar Class IX](#)[NCERT Exemplar Class X](#)[NCERT Exemplar Class XI](#)**Unit Name**

Unit 4: Numerical Reasoning and Probability

Description

This unit builds on the first unit by using arithmetic to understand crucial ideas applicable in everyday life. This includes working with ratios and proportions, percentages, simple and compound interest, etc. Focus will be on the simplest of relations when two quantities are related linearly.

The unit will also touch upon some important concepts like progression, permutation, combination, and probability.

Week 5:

4.1 Percentage

4.2 Ratio and Proportion

i) Direct Proportion

ii) Inverse Proportion

4.3 Arithmetic Progression

Week 6:
4.4 Permutation and Combination
4.5 Probability

Readings

Core Readings:

NCERT Class VI, Chapter:

- Ratio and Proportion

NCERT Class VII, Chapter:

- Comparing Quantities

NCERT Class VIII, Chapters:

- Comparing Quantities
- Direct and Inverse Proportions

NCERT Class IX, Chapter:

- Probability

NCERT Class X, Chapters:

- Arithmetic Progression
- Probability

NCERT Class XI, Chapters:

- Probability
- Permutation and Combination

All core readings for this chapter are uploaded [here](#).

Supplementary Readings:

[NCERT Exemplar Class IX](#)

[NCERT Exemplar Class X](#)

[NCERT Exemplar Class XI](#)

Annexure

ILO Assessment Linkage

ILO	Assessment	Assessment Notes
Demonstrate	In-class exam	Exam 1 (60 minutes duration, to be submitted individually), to be conducted in 2nd week, will be a pen and paper-based assessment on the syllabus of unit 1. The assessment will include a mix of question types (multiple-choice and short problem-solving questions) and will evaluate students' foundational understanding of the quantitative methods.
	In-class exam	Exam 2 (60 minutes duration, to be submitted individually), to be conducted in 3rd week, will be a pen and paper-based assessment on the syllabus of unit 2. The assessment will include a mix of question types (multiple-choice and short problem-solving questions) and will evaluate students' ability to understand algebra and coordinate geometry.
	Class participation	This will be based on the students’ interactions with peers and their contribution to discussions in the class. The students will be divided into groups and engaged in different types of in-class activities. The instructor may use this assessment to gather continuous feedback from the students, including their challenges in navigating the course, and take the necessary actions.
		Exam 4 (60 minutes duration, to be submitted individually), to be conducted in 6th week, will be a pen and paper-based assessment on the syllabus of unit 4. The assessment will include a mix of question types (multiple-choice and short problem-solving questions) and will evaluate students' ability to apply quantitative knowledge in everyday life.

Engage	In-class exam Project Class participation	The students will be divided into groups. Each group will work on a project related to an everyday life issue (for example, breakfast habit, reading time). This will involve data acquisition (from small surveys), processing, and visualization. The project will start in the 3rd week of the course. Each group will present their project findings as a digital/poster presentation. The presentation will carry 10% weightage. It will be evaluated based on topic selection and presentation skill (including visual appeal and interaction). Individual members must indicate their roles in the project. Individual student will submit a brief report, which will contribute 10% to the overall grade. The report will be evaluated based on its organization and information of the workflow (data collection, analysis, and interpretation of the findings). The viva-voce will evaluate contribution and depth of knowledge of individual group members and will contribute 10% to the total grade. In the viva-voce, the instructor may ask the students to demonstrate use of a specific tool or function of the software related to their projects. The instructor should ensure that the groups are heterogeneous with students across all disciplines. The teams should not be larger than 5 and not smaller than 2. All projects will be discussed a priori. The students should start working on the projects from week 3. The presentation, report submission, and viva-voce will be conducted in week 5. This will be based on the students’ interactions with peers and their contribution to discussions in the class. The students will be divided into groups and engaged in different types of in-class activities. The instructor may use this assessment to gather continuous feedback from the students, including their challenges in navigating the course, and take the necessary actions.
Execute	In-class exam Project Class participation	Exam 3 (60 minutes duration, to be submitted individually), to be conducted in 4th week, will be a computer-based assessment on the syllabus of Unit 3. The assessment will evaluate students' ability to estimate statistical measures in a spreadsheet program. The students will be divided into groups. Each group will work on a project related to an everyday life issue (for example, breakfast habit, reading time). This will involve data acquisition (from small surveys), processing, and visualization. The project will start in the 3rd week of the course. Each group will present their project findings as a digital/poster presentation. The presentation will carry 10% weightage. It will be evaluated based on topic selection and presentation skill (including visual appeal and interaction). Individual members must indicate their roles in the project. Individual student will submit a brief report, which will contribute 10% to the overall grade. The report will be evaluated based on its organization and information of the workflow (data collection, analysis, and interpretation of the findings). The viva-voce will evaluate contribution and depth of knowledge of individual group members and will contribute 10% to the total grade. In the viva-voce, the instructor may ask the students to demonstrate use of a specific tool or function of the software related to their projects. The instructor should ensure that the groups are heterogeneous with students across all disciplines. The teams should not be larger than 5 and not smaller than 2. All projects will be discussed a priori. The students should start working on the projects from week 3. The presentation, report submission, and viva-voce will be conducted in week 5. This will be based on the students’ interactions with peers and their contribution to discussions in the class. The students will be divided into groups and engaged in different types of in-class activities. The instructor may use this assessment to gather continuous feedback from the students, including their challenges in navigating the course, and take the necessary actions.
Perform	In-class exam In-class exam In-class exam Class participation	Exam 1 (60 minutes duration, to be submitted individually), to be conducted in 2nd week, will be a pen and paper-based assessment on the syllabus of unit 1. The assessment will include a mix of question types (multiple-choice and short problem-solving questions) and will evaluate students' foundational understanding of the quantitative methods. Exam 2 (60 minutes duration, to be submitted individually), to be conducted in 3rd week, will be a pen and paper-based assessment on the syllabus of unit 2. The assessment will include a mix of question types (multiple-choice and short problem-solving questions) and will evaluate students' ability to understand algebra and coordinate geometry. Exam 4 (60 minutes duration, to be submitted individually), to be conducted in 6th week, will be a pen and paper-based assessment on the syllabus of unit 4. The assessment will include a mix of question types (multiple-choice and short problem-solving questions) and will evaluate students' ability to apply quantitative knowledge in everyday life. This will be based on the students’ interactions with peers and their contribution to discussions in the class. The students will be divided into groups and engaged in different types of in-class activities. The instructor may use this assessment to gather continuous feedback from the students, including their challenges in

navigating the course, and take the necessary actions.

Assessment ILO Linkage

Assessment	Assessment Notes	ILO
Class participation	This will be based on the students' interactions with peers and their contribution to discussions in the class. The students will be divided into groups and engaged in different types of in-class activities. The instructor may use this assessment to gather continuous feedback from the students, including their challenges in navigating the course, and take the necessary actions.	Demonstrate Perform Execute Engage
In-class exam	Exam 1 (60 minutes duration, to be submitted individually), to be conducted in 2nd week, will be a pen and paper-based assessment on the syllabus of unit 1. The assessment will include a mix of question types (multiple-choice and short problem-solving questions) and will evaluate students' foundational understanding of the quantitative methods.	Demonstrate Perform
In-class exam	Exam 2 (60 minutes duration, to be submitted individually), to be conducted in 3rd week, will be a pen and paper-based assessment on the syllabus of unit 2. The assessment will include a mix of question types (multiple-choice and short problem-solving questions) and will evaluate students' ability to understand algebra and coordinate geometry.	Demonstrate Perform
In-class exam	Exam 3 (60 minutes duration, to be submitted individually), to be conducted in 4th week, will be a computer-based assessment on the syllabus of Unit 3. The assessment will evaluate students' ability to estimate statistical measures in a spreadsheet program.	Execute
In-class exam	Exam 4 (60 minutes duration, to be submitted individually), to be conducted in 6th week, will be a pen and paper-based assessment on the syllabus of unit 4. The assessment will include a mix of question types (multiple-choice and short problem-solving questions) and will evaluate students' ability to apply quantitative knowledge in everyday life.	Perform Engage
Project	The students will be divided into groups. Each group will work on a project related to an everyday life issue (for example, breakfast habit, reading time). This will involve data acquisition (from small surveys), processing, and visualization. The project will start in the 3rd week of the course. Each group will present their project findings as a digital/poster presentation. The presentation will carry 10% weightage. It will be evaluated based on topic selection and presentation skill (including visual appeal and interaction). Individual members must indicate their roles in the project. Individual student will submit a brief report, which will contribute 10% to the overall grade. The report will be evaluated based on its organization and information of the workflow (data collection, analysis, and interpretation of the findings). The viva-voce will evaluate contribution and depth of knowledge of individual group members and will contribute 10% to the total grade. In the viva-voce, the instructor may ask the students to demonstrate use of a specific tool or function of the software related to their projects. The instructor should ensure that the groups are heterogeneous with students across all disciplines. The teams should not be larger than 5 and not smaller than 2. All projects will be discussed a priori. The students should	Execute Engage

	start working on the projects from week 3. The presentation, report submission, and viva-voce will be conducted in week 5.	
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